

Human Academic Authorship Reasoning (HAAR)

Framework v1.2

Evidence-Based Assessment of Authorial Command of Scientific Reasoning

Version 1.2 introduces assessment-instrument validation mechanisms, including the Answer-Position Neutrality Principle, option permutation, positional audit, and cross-check independence.

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1. Purpose

HAAR — Human Academic Authorship Reasoning is a framework for evidence-based assessment of how well a person claiming authorship of a scientific or academic work commands the reasoning, decisions, assumptions, alternatives, methodological transitions, interpretations, and limitations underlying that work.

HAAR does not attempt to determine authorship from linguistic style, textual similarity, or the presumed human/AI origin of the wording.

Its primary question is:

Does the respondent demonstrate stable and transferable command of the critical scientific reasoning transitions underlying the submitted work?

HAAR produces a structured reasoning evidence profile. The final judgment concerning authorship, scientific contribution, or responsibility remains a human decision.

HAAR → EVIDENCE

HUMAN → DECISION

HAAR does not output an “AI percentage” or an authorship probability.

$S_{\text{HAAR}} \neq P(\text{authorship})$

2. Fundamental HAAR Rule

HAAR must not primarily ask:

“Who generated this text?”

It must ask:

“Does the respondent command the scientific reasoning represented by this work, including the ability to reconstruct, justify, challenge, modify, and extend its critical transitions?”

Text Authorship ≠ Reasoning Ownership ≠ Scientific Responsibility

Text provenance may constitute additional evidence, but it is not equivalent to demonstrated command of scientific reasoning.

3. Core Principles

1. Reasoning over Style — assess scientific reasoning rather than linguistic fingerprints.
2. Edge-Centric Assessment — assess transitions between reasoning elements rather than isolated facts.
3. Milestone-Based Questioning — concentrate assessment on scientifically consequential transitions.
4. Work-Specific Personalization — derive questions from the submitted work.
5. Plausible Alternatives — use scientifically credible distractors and alternative reasoning branches.
6. Transformation over Paraphrase — require reasoning under modified conditions rather than textual reproduction.

7. Anti-Coaching Principle — do not teach the expected reasoning during active assessment.
8. Delayed Cross-Check — revisit important transitions later using a different reasoning operation.
9. Evidence Localization — map evidence back to specific reasoning transitions.
10. Human Decision Principle — HAAR provides evidence; a human makes the final decision.
11. Answer-Position Neutrality Principle — the position of the target answer must not provide information about correctness.

The eleventh principle is introduced in HAAR v1.2 following experimental application of the framework.

4. Input

$$T = \{t_1, t_2, \dots, t_n\}$$

Possible inputs include:

- scientific article
- conference paper
- thesis
- bachelor project
- master's thesis
- PhD dissertation
- monograph
- research report
- project proposal
- other structured research work

Optional supplementary evidence may include:

- experimental protocols
- source code
- datasets
- laboratory records
- preregistrations
- research notes
- version histories
- rejected hypotheses
- failed experiments
- correspondence concerning methodological decisions
- reproducibility packages

These artifacts strengthen provenance assessment but do not replace the reasoning assessment.

5. Semantic Reasoning Network

HAAR reconstructs a Semantic Reasoning Network:

$$G_R = (V, E, \tau_V, \tau_E, w)$$

The network represents the scientific reasoning encoded in the work.

5.1 Node Types

$$\tau_V(v) \in \{P, A, H, M, D, E_x, O, I, C, R, L, F\}$$

- P — PROBLEM
- A — ASSUMPTION
- H — HYPOTHESIS
- M — METHOD

- D — DATA
- E_x — EXPERIMENT
- O — OBSERVATION
- I — INTERPRETATION
- C — CLAIM
- R — RESULT
- L — LIMITATION
- F — FUTURE_STEP

A typical reasoning trajectory may be represented as:

$$P \rightarrow H \rightarrow M \rightarrow E_x \rightarrow O \rightarrow I \rightarrow R$$

6. Reasoning Edge Types

Edges represent scientific reasoning transitions. Possible edge types include:

- MOTIVATES
- ASSUMES
- SELECTS
- USES
- TESTS
- SUPPORTS
- CONTRADICTS
- EXPLAINS
- IMPLIES
- DEPENDS_ON
- LIMITS
- GENERALIZES
- LEADS_TO

Not every edge should be interpreted as a strict causal relation. The network is therefore a scientific reasoning network, not necessarily a purely causal graph.

7. Hidden Transitions

Scientific texts frequently compress important reasoning steps. A visible transition

$$A \rightarrow B$$

may actually correspond to

$$A \rightarrow \{X_1, X_2, \dots, X_n\}_{\text{hidden}} \rightarrow B$$

$$H_{ij} = \{X_1, \dots, X_n\}$$

Hidden transitions may include:

- rejected alternatives
- implicit assumptions
- abandoned hypotheses
- failed experiments
- selection criteria
- methodological compromises
- reasons for rejecting another method
- interpretation choices
- unexpected observations

These transitions are especially informative because they are often difficult to recover through simple textual memorization.

8. Milestone Detection

HAAR identifies milestones — transitions or subnetworks whose understanding is particularly important for demonstrating command of the work.

$$K(m_i) = \alpha C_i + \beta D_i + \gamma N_i + \delta U_i$$

- C_i — structural importance
- D_i — dependency of important conclusions
- N_i — non-triviality
- U_i — uncertainty, alternatives, or decision complexity

$$M^* = \text{TopK}_{\{m_i\}} K(m_i)$$

Structural centrality alone is insufficient: a peripheral but highly consequential methodological choice may be more informative than a structurally central but trivial transition.

9. Critical Transition Identification

$$e_{ij} = (v_i \rightarrow v_j)$$

Assessment should focus primarily on transitions rather than isolated nodes.

$$H \rightarrow M \quad M \rightarrow E_x \quad E_x \rightarrow O \quad O \rightarrow I \quad I \rightarrow R$$

The system should determine what scientific decision or inference connects the two nodes.

10. Personalized Questionnaire Generation

$$q_i = (s_i, O_i)$$

- s_i — question stem
- O_i — answer options

$$O_i = \{o_{i1}, o_{i2}, \dots, o_{ik}\}, \quad 4 \leq k \leq 6$$

$$|\text{Target}(q_i)| = 1$$

Questions must be derived from the reconstructed reasoning network rather than from generic disciplinary knowledge whenever possible.

11. Question Types

WHY — Why was transition $A \rightarrow B$ scientifically justified?

WHY-NOT — Why was alternative B' not selected?

ALTERNATIVE — What would constitute a plausible alternative to B ?

COUNTERFACTUAL — What changes if assumption A no longer holds?

REMOVE — What happens to the reasoning structure if node or transition X is removed?

CONTRADICT — How should the reasoning change if new evidence contradicts an existing assumption or observation?

LIMIT — Under what conditions does the conclusion cease to be justified?

TRANSFER — Can the reasoning be transferred to another domain, dataset, scale, or research context?

EXTEND — What scientifically justified next research transition follows from the current network?

REPRODUCE — What must be reproduced to verify the corresponding result?

A strong HAAR assessment should combine several of these operations.

12. Distractor Construction

Incorrect alternatives must be scientifically plausible. Distractors may be generated from:

- neighboring reasoning branches
- rejected hypotheses
- alternative methods
- superficially plausible interpretations
- incorrect dependency directions
- overgeneralized conclusions
- plausible but unsupported causal claims

Each option may be internally tagged as:

TARGET | PLAUSIBLE | TEXTUAL | ALTERNATIVE | CONTRADICTORY | OVERCLAIM

These labels must remain hidden from the respondent.

13. Question Validation

Context validity. The question follows from the submitted work or an explicitly stated perturbation.

Target uniqueness. $|\text{Target}(q_i)| = 1$.

Distractor plausibility. Incorrect options must remain credible enough to require scientific reasoning.

Semantic independence. The correct answer should not be identifiable merely by lexical overlap with the article.

No unsupported external dependency. The question should not require knowledge absent from the work unless explicitly introduced as part of a perturbation.

No trivial textual retrieval. The question should preferably require reasoning rather than sentence matching.

No answer-length leakage. The target answer should not systematically be longer, more detailed, or linguistically more sophisticated than distractors.

No position leakage. The location of the target option must not form a detectable pattern.

14. Answer-Position Neutrality Principle

New in HAAR v1.2.

The position of the target answer must not provide evidence about which answer is correct.

$$\pi(q_i) \in \{A, B, C, D, E, \dots\}$$

For a questionnaire Q with k answer positions, define:

$$p_j = |\{q_i : \pi(q_i) = j\}| / |Q|$$

For sufficiently large questionnaires:

$$p_j \approx 1/k$$

For five-option questions:

$$p_A \approx p_B \approx p_C \approx p_D \approx p_E \approx 0.20$$

$$\Delta_{\text{pos}} = \max_j p_j - \min_j p_j$$

$$\Delta_{\text{pos}} \leq \epsilon_{\text{pos}}$$

Simple deterministic rotation is prohibited. The sequence A, B, C, D, E, A, B, C, D, E, ... is balanced but predictable and therefore violates the principle.

POSITIONAL BALANCE + POSITIONAL UNPREDICTABILITY

15. Answer-Position Audit

$$\Pi_Q = [\pi(q_1), \pi(q_2), \dots, \pi(q_n)]$$

The audit checks for:

- excessive concentration in one position
- long runs of the same target position
- deterministic cycles
- alternating patterns
- correlation between question type and answer position
- correlation between question difficulty and answer position
- correlation between milestone type and answer position

A questionnaire failing the positional audit must be reshuffled and revalidated before use.

16. Option Permutation

$$q_i^{\text{semantic}} \rightarrow \text{Validate} \rightarrow \text{PermuteOptions} \rightarrow q_i^{\text{display}}$$

The target is determined before option positions are assigned.

$$\text{PermuteOptions}(q_i) \not\Rightarrow \text{semantic modification}$$

After permutation, target uniqueness must be checked again. This prevents positional design decisions from contaminating semantic question construction.

17. Sequential Adaptive Interview

$$q_1 \rightarrow a_1 \rightarrow q_2(a_1) \rightarrow a_2 \rightarrow \dots \rightarrow q_n$$

The next question may depend on previous evidence. However, adaptation must concern reasoning content, not reveal whether the previous answer was correct.

18. Anti-Coaching Principle

$$\text{ANSWER} \rightarrow \text{RECORD} \rightarrow \text{NEXT QUESTION}$$

and not

$$\text{ANSWER} \rightarrow \text{EXPLANATION} \rightarrow \text{HINT} \rightarrow \text{NEXT QUESTION}$$

The respondent should not receive correctness feedback, explanatory hints, or the expected reasoning while the assessment is still active. Otherwise the system risks measuring reasoning taught by HAAR rather than reasoning already commanded by the respondent.

19. Delayed Cross-Check

A critical transition should not normally be classified from one response.

$$q_j = \Phi(q_i, e), \quad j > i + d, \quad d > 0$$

The later question should use a different reasoning operation. Immediate paraphrasing of the same question is not a proper Delayed Cross-Check.

20. Cross-Check Independence

New in HAAR v1.2.

$$D(q_i, q_j) \geq D_{\min}$$

Functional differences may involve:

- different reasoning operation
- different perturbation
- reversed reasoning direction
- alternative methodological branch
- new contradictory observation
- removal of an assumption
- transfer to another context

This prevents pseudo-cross-checking through superficial paraphrase.

21. Response Recording

$$a_i = (q_i, m_i, e_i, o_i, t_i, c_i)$$

- q_i — question
- m_i — associated milestone
- e_i — tested transition
- o_i — selected option
- t_i — question type
- c_i — contextual metadata

For audit purposes, HAAR should also record the displayed option permutation ρ_i .

22. Reasoning Evidence Graph

$$G_E = (V, E^+, E?, E^-)$$

- E^+ — supported or stable transitions
- $E?$ — insufficient, uncertain, or mixed evidence
- E^- — repeatedly contradicted transitions

A single incorrect response should not normally move a transition directly into E^- . Evidence classification should consider repeated assessment, question diversity, delayed cross-checks, counterfactual performance, consistency across alternative formulations, and criticality of the tested transition.

23. Local Evidence Aggregation

$$S(m_i) = (\sum_j w_{ij} s(q_{ij})) / (\sum_j w_{ij})$$

Question weights may depend on milestone criticality, reasoning-operation difficulty, independence of the question, cross-check value, and transformation depth. A simple recognition question should not necessarily carry the same evidential weight as a successful counterfactual transformation.

24. Global Evidence Aggregation

$$S_{\text{HAAR}} = (\sum_i K(m_i) S(m_i)) / (\sum_i K(m_i))$$

$$S_{\text{HAAR}} \neq P(\text{authorship})$$

For example, $S_{\text{HAAR}} = 0.91$ means high consistency with the tested reasoning structure under the adopted scoring model. It does not mean “91% probability of authorship.”

25. Structural Evidence Dominates Raw Score

Two respondents may have identical aggregate scores while exhibiting substantially different reasoning evidence.

$$S_HAAR^{(1)} = S_HAAR^{(2)}$$

One may show uncertainty on peripheral transitions, while another shows contradictions on high-criticality transitions. Therefore S_HAAR must always be interpreted together with G_E . The Evidence Graph is not merely a visualization of the aggregate score; it is a primary assessment artifact.

26. Evidence Profile

- SUPPORTED TRANSITIONS
- UNCERTAIN TRANSITIONS
- CONTRADICTORY TRANSITIONS
- COUNTERFACTUAL PERFORMANCE
- ALTERNATIVE REASONING
- STRUCTURAL PERTURBATION PERFORMANCE
- REASONING CONSISTENCY
- DELAYED CROSS-CHECK RESULTS
- ANSWER-POSITION AUDIT
- OVERALL REASONING EVIDENCE

The answer-position audit concerns the quality of the assessment instrument, not the respondent's scientific competence.

27. Evidence Categories

- STRONG CONSISTENCY
- CONSISTENT
- PARTIALLY CONSISTENT
- INSUFFICIENT EVIDENCE
- SIGNIFICANT INCONSISTENCIES

These categories describe the observed reasoning evidence. They must not be presented as automatic declarations of historical authorship.

28. Human Decision Layer

$$R_HAAR = \{E^+, E?, E^-, C_f, A_r, D_c, P_a\}$$

- C_f — counterfactual performance
- A_r — alternative reasoning
- D_c — delayed cross-check results
- P_a — positional audit

The human assessor may additionally consider research provenance, source files, experiment histories, laboratory records, version histories, source code, raw data, institutional requirements, oral defense, and scientific responsibility.

$$R_HAAR \rightarrow \text{Human Assessment} \rightarrow \text{Final Decision}$$

29. Complete Operational Cycle

INPUT → PARSE → REASONING NETWORK → MILESTONES → CRITICAL TRANSITIONS → HIDDEN TRANSITIONS → QUESTION GENERATION → DISTRACTOR GENERATION → QUESTION VALIDATION → OPTION PERMUTATION → POSITION AUDIT → QUESTION ORDERING → INTERVIEW → ADAPT →

DELAYED CROSS-CHECK → EVIDENCE MAPPING → EVIDENCE GRAPH → AGGREGATION → REPORT →
HUMAN DECISION

QUESTION SEMANTICS ≠ ANSWER POSITION

30. Abstract Operational Example

$H_1 \rightarrow M_1 \rightarrow E_1 \rightarrow O_1 \rightarrow I_1 \rightarrow R_1$

Suppose $M_1 \rightarrow E_1$ is identified as a critical transition. HAAR may first ask why E_1 was appropriate for evaluating M_1 ; later ask what changes if M_1 is unavailable; and later introduce a new observation O_1 that contradicts an assumption required by I_1 .

WHY → REMOVE → CONTRADICT

The correct options may appear, for example, in positions D, A, and E, but these positions are assigned independently of semantics and checked globally for positional leakage.

31. Education and Qualification Levels

Coursework → Bachelor → Master → PhD → Scientific Article → Dissertation

The architecture remains stable, while the depth of reasoning operations changes. At higher qualification levels, greater weight should be given to WHY-NOT, ALTERNATIVE, COUNTERFACTUAL, CONTRADICT, LIMIT, REPRODUCE, and EXTEND.

For doctoral assessment, the ability to defend, revise, and extend reasoning is more informative than textual recall.

32. Relationship Between HAAR and HAAG

HAAG and HAAR may share a common representation of scientific reasoning but perform opposite functions.

HAAG: Human → Scientific Reasoning → Research Process → Scientific Text

HAAR: Scientific Text → Scientific Reasoning Network → Assessment → Human Evidence

HAAG: Reasoning → Text

HAAR: Text → Reasoning → Evidence

HAAR remains independent of whether HAAG, another agentic system, an LLM, or conventional writing tools were used to produce the submitted work.

33. Limitations

HAAR cannot directly observe internal cognition. It cannot establish from questionnaire performance alone exactly who historically created every part of a scientific work. A knowledgeable non-author may understand a work deeply.

A genuine author may perform poorly because of language difficulties, anxiety, disability, memory limitations, time elapsed since the research, or differences between written and oral performance.

Poor reconstruction of G_R may generate poor questions. Poorly designed distractors may distort results. Predictable target positions may artificially inflate performance. Therefore, high-stakes use requires empirical calibration.

34. Empirical Validation

G_A = Authors

G_K = Knowledgeable Non-Authors

G_I = Independent Readers

Validation should examine more than total accuracy. Useful measures include:

- WHY performance
- WHY-NOT performance
- counterfactual performance
- alternative reasoning
- structural perturbation
- delayed cross-check consistency
- $E_{\square}/E_{?}/E_{\square}$ distributions
- milestone-specific evidence
- positional bias sensitivity

An especially important ablation is:

HAAR_neutral vs. HAAR_biased-position

This can quantify how much apparent performance is produced by positional leakage rather than reasoning command.

35. HAAR v1.2 Output Package

SOURCE_WORK
 SEMANTIC_REASONING_NETWORK
 MILESTONE_LIST
 CRITICAL_TRANSITIONS
 HIDDEN_TRANSITIONS
 QUESTIONNAIRE
 QUESTION_TYPES
 ANSWER_OPTIONS
 QUESTION_TRANSITION_MAP
 EXPECTED_REASONING_MAP
 OPTION_PERMUTATION_LOG
 ANSWER_POSITION_DISTRIBUTION
 ANSWER_POSITION_AUDIT
 HUMAN_ANSWERS
 ADAPTIVE_QUESTION_PATH
 DELAYED_CROSS_CHECK_MAP
 ANSWER_EVIDENCE
 REASONING_EVIDENCE_GRAPH
 SUPPORTED_TRANSITIONS
 UNCERTAIN_TRANSITIONS
 CONTRADICTORY_TRANSITIONS
 COUNTERFACTUAL_PERFORMANCE
 ALTERNATIVE_REASONING
 STRUCTURAL_PERTURBATION_PERFORMANCE
 AGGREGATED_EVIDENCE
 OVERALL_REASONING_EVIDENCE
 HAAR_ASSESSMENT

HUMAN_DECISION

The addition of OPTION_PERMUTATION_LOG, ANSWER_POSITION_DISTRIBUTION, and ANSWER_POSITION_AUDIT makes positional neutrality auditable rather than merely declarative.

36. Fundamental HAAR v1.2 Rule

HAAR must not ask primarily:

“Did this person write this text?”

Nor should it ask:

“Can this person reproduce what the text says?”

Its primary operational question is:

“Can this person reconstruct, justify, challenge, transform, and extend the critical scientific reasoning represented by the work?”

The assessment itself must simultaneously satisfy:

SEMANTIC VALIDITY + REASONING DIVERSITY + CROSS-CHECK INDEPENDENCE + POSITIONAL
NEUTRALITY

The final invariant remains:

HAAR → EVIDENCE

HUMAN → DECISION